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Roll No

CE-8004(3)-CBGS

B.E. VIII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Geo-Informatics

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1.
 - a) What do you understand by the term GIS? Describe the concept of GIS in detail.
 - b) Explain spatial data modeling and data output in a GIS database.
2. The following data is given for flight planning:
 - i) Size of photo 18 cm × 18 cm
 - ii) Focal length 21 cm
 - iii) Scale 1:10000
 - iv) Longitudinal overlap 60%
 - v) Lateral overlap 30%
 - vi) Terrain length 450 km
 - vii) Terrain width 360 km
 - viii) Ground speed of aircraft 250 km/hCalculate the following:
 - i) Elevation of the aircraft
 - ii) Number of flight strips
 - iii) Number of photos per strips
 - iv) Exposure interval
 - v) Number of photographs

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3.
 - a) Discuss the role of Geo-informatics in urban growth modeling.
 - b) Discuss about High Resolution Remote Sensing.
4. Describe the GIS and Digital Image Processing techniques are helpful to do accident analyses effectively?
5.
 - a) Explain the effects of Orientation elements in Photogrammetry.
 - b) Write about elements of Aero Triangulation and Analytical Method.
6.
 - a) Explain various Geometric and radiometric corrections to satellite data.
 - b) An image of a hill top is 88.5 mm from the centre of a photograph. The elevation of the hill is 665m and the flight altitude is 5000m from the same datum. How much is the image displaced use to elevation of the hill?
7.
 - a) Explain about GPS Receivers and GPS observation methods.
 - b) What are the advantages over conventional methods of GPS observation methods?
8. Define the following:
 - i) Terrestrial Photogrammetry
 - ii) Satellite based Surveys
 - iii) Stereo plotters
 - iv) Web GIS
